

StochLean Markov Processes, Transition Systems, Recurrence, Coupling & Feller Design v0.1

Normative Work implementation handoff | Klenke Ch. 14.4, 17, 18, and 21.26-21.27

Status: DESIGN FROZEN FOR IMPLEMENTATION

This document is a functional ownership and implementation contract, not a chapter-by-chapter port. Klenke chapter/section numbers are provenance and coverage references only. Every public declaration remains subject to the global duplicate audit and trust-boundary rules before construction.

0. Global implementation contract

The package must obey the project-wide no-duplicate hard rule: before any new public declaration, Work searches the active/pinned Mathlib revision, then the complete local StochLean tree, then the approved-external ledger. If an exact or better canonical implementation exists, reuse or bridge it; do not mirror it locally. Compilation is not sufficient: statement, semantic, dependency, axiom, trust-boundary, minimal-import, and source audits are mandatory.

Build order is functional. Do not organize implementation around fixed chapter blocks. The canonical process representation remains $X : T \rightarrow \Omega \rightarrow E$; do not create a StochasticProcess or MarkovProcess bundle merely to package existing functions, kernels, filtrations, and laws.

0.1 External-proof protocol - normative for this and future packages

For textbook theorems whose full proof is not supplied in Klenke, Work must first follow the textbook citation to a concrete public source. The handoff must give the source name and exact chapter/proposition/theorem pointer whenever known. If the cited proof is publicly accessible, Work audits the exact hypotheses, conclusion, proof dependencies, and correctness. If that audit passes, formalize the audited proof (or the minimal dependency closure it genuinely needs) and add it to StochLean under the canonical owner. If the cited material cannot be accessed in sufficient detail, mark the theorem DEFER with the inaccessible source pointer and move on. Do not launch an independent proof attempt merely to close the textbook theorem.

This policy is deliberate: standard textbook results are presumed mathematically established; the goal is a reliable Lean library, not rediscovery. Independent proof search is allowed only when explicitly requested later or when no external proof is needed because Mathlib already contains a reusable theorem.

0.2 Maintenance ledger - mandatory format

Every large package report must include a Blocker & Deferred-Dependency Ledger. Each row must end in RESOLVED, STILL-BLOCKED, or DEFERRED (owner + reason), and must identify downstream impact. This format is now a standing maintenance rule for subsequent packages as well.

ID	Content	Owner	Status / reason	Unlock condition / downstream
MK-B01	StandardBorel arbitrary-index KET external audit inherited from prior package	Projective construction	Inherited blocker if not yet closed	Audit/bridge complete; unlock canonical Markov-process construction
MK-B02	futureMeasurableSpace	Generic Process / Information Flow	Mandatory current dependency	Implement/reuse before conditional-independence characterization
MK-B03	bounded Q-matrix $\Rightarrow$ unique semigroup	Markov/Generator	Current theorem blocker	Uniformization existence + rigorous uniqueness
MK-B04	Klenke-Feller $\Rightarrow$ C0 sup-norm strong continuity	Markov/Feller	Mandatory bridge	Audit known proof; formalize equivalence
MK-B05	Klenke 21.27 cited proof audit	External proof audit	External-proof blocker	Access and audit Rogers-Williams / Revuz-Yor cited proof
MK-B06	RCLL/càdlàg Feller realization prerequisites	Canonical path/regularization owner	Conditional prerequisite	Formalize only the audited proof dependency closure
MK-B07	right-continuous/completed/usual filtration prerequisites	Generic Filtration owner	Conditional prerequisite	Do not duplicate inside Markov
MK-B08	strong-Markov stopping-time closure	Markov + stopping-time interfaces	Blocker after B05-B07	Finish audited 21.27 chain
MK-D01	general Banach generator theory	Future semigroup/SDE layer	DEFERRED	Not needed for v0.1 Markov completion
MK-D02	full Skorokhod D topology	Functional weak-convergence package	DEFERRED	Owned by future path-space package

MK-D03	MCMC/Metropolis application layer	Future application package	DEFERRED	Core invariant/reversible/convergence first
MK-D04	finite-state spectral convergence rates	Future finite-state spectral layer	DEFERRED	Not a core blocker

1. Scope and ownership

Primary textbook provenance: Klenke §14.4 (consistent transitions and Markov semigroups), Ch. 17 (Markov chains/processes, strong Markov in countable time, countable-state recurrence, invariant measures, continuous-time chains/Q-matrices), Ch. 18 (periodicity, coupling, convergence to equilibrium), and §21.26-21.27 (Feller semigroups and strong-Markov RCLL realization).

Core package scope is: information-flow future sigma-fields; general two-parameter TransitionSystem; homogeneous MarkovSemigroup; kernel-free Markov semantic property; transition-system/semigroup representations; canonical path laws; countable-chain reachability, hitting, recurrence, Green function, invariant probability and Kac formula; periodicity; generic probability couplings and total-variation bounds; independent coalescent and TV convergence; bounded Q-matrices and uniformization; Feller operator interface and strong-Markov realization route.

Not owned here: generic Kernel definitions/composition, condExp/condDistrib, filtrations/stopping times, projective/Kolmogorov extension, full path-space topology, general Banach semigroup generators, or SDE/martingale-problem theory.

2. Information-flow prerequisite: future sigma-field

Work must expose the generic process concept futureMeasurableSpace X_t , with mathematical meaning $F^{\wedge}\{X,+\}_t = \sigma(X_s : s \geq t)$. It belongs under generic Process/Filtration or Process/InformationFlow ownership, not under a Markov-private namespace.

futureMeasurableSpace X_t – provisional public name; duplicate audit first

Required API: $\sigma(X_t) \leq F^{\wedge}\{X,+\}_t$; antitonicity $t \leq u \Rightarrow F^{\wedge}\{X,+\}_u \leq F^{\wedge}\{X,+\}_t$; and minimality: if every X_s for $s \geq t$ is G -measurable, then $F^{\wedge}\{X,+\}_t \leq G$. For discrete time, bridge the existing canonical tail sigma-field by $T^{\wedge}X = \bigcap_n F^{\wedge}\{X,+\}_n$; do not redefine the tail sigma-field.

3. Kernel orientation - mandatory Work warning

Mathlib defines $\eta \circ_k \kappa$ to mean first run κ , then run η : $(\eta \circ_k \kappa)(x,A) = \int \eta(y,A) \kappa(x,dy)$. Traditional probability notation writes chronological kernel multiplication as $K_{\{r,s\}} K_{\{s,t\}}$. Therefore the probability identity $K_{\{r,s\}} K_{\{s,t\}} = K_{\{r,t\}}$ is written in Lean as $K \ r \ t = K \ s \ t \circ_k K \ r \ s$.

Probability notation	Mathlib/StochLean Lean notation
$K_{\{r,s\}} K_{\{s,t\}}$	$K \ s \ t \circ_k K \ r \ s$
first $K_{\{r,s\}}$, then $K_{\{s,t\}}$	$K \ s \ t \circ_k K \ r \ s$
$K_{\{r,s\}} K_{\{s,t\}} = K_{\{r,t\}}$	$K \ r \ t = K \ s \ t \circ_k K \ r \ s$
homogeneous $K_s K_t = K_{s+t}$	$K \ (s+t) = K \ t \circ_k K \ s$
operator action $P_{\{s+t\}} f$	$P \ s \ (P \ t \ f)$

StochLean definitions use the Mathlib $\circ_k$ orientation. Do not define a second probability-order multiplication operator. The Work push report must repeat this orientation warning so future maintainers do not reverse Chapman-Kolmogorov or Feller operator composition.

4. General-first transition architecture

4.1 TransitionSystem - general two-parameter object

The two-parameter non-time-homogeneous evolution is named TransitionSystem, not MarkovSemigroup. The public property should be Prop-valued (for example IsMarkovTransitionSystem or equivalent final audited name), with K defined only on legal forward-time pairs rather than totalizing meaningless $r > s$ cases.

$K : \forall r \ s : T, r \leq s \rightarrow \text{Kernel } E \ E$

Core assumptions can start from Preorder T . Required laws: each $K\ r\ s$ hrs is a Markov kernel; $K_t\ t = \text{Kernel.id}$; and for $r \leq s \leq t$, $K_r\ t = K_s\ t \circ_k K_r\ s$. Klenke §14.42 supplies the consistent-family source; the explicit diagonal identity is the natural canonical completion and must be marked as such in source mapping.

4.2 MarkovSemigroup - homogeneous one-parameter object

The homogeneous object is named `MarkovSemigroup` and should again be a property of $\kappa : T \rightarrow \text{Kernel } E\ E$, not a heavyweight structure containing the kernels. With `AddCommMonoid T`, require $\kappa\ 0 = \text{Kernel.id}$, every $\kappa\ t$ is Markov, and $\kappa(s+t) = \kappa\ t \circ_k \kappa\ s$. Commutativity of time addition then yields $\kappa\ s \circ_k \kappa\ t = \kappa\ t \circ_k \kappa\ s$.

Discrete sanity: $n \mapsto \kappa^n n$ must instantiate `MarkovSemigroup` and agree exactly with `Mathlib Kernel.pow_add`. The general `TransitionSystem` is built first; homogeneous references explicitly state when they use `MarkovSemigroup`.

4.3 Homogeneous-to-general bridge

Do not define $K_s\ t = \kappa(t-s)$ on an arbitrary ordered additive type using truncated subtraction. Provide this bridge only for time types with a mathematically valid forward difference, such as `Nat` or `NNReal` / appropriate ordered additive structures. Most downstream recurrence and Feller results use the homogeneous object directly and do not need an over-generalized subtraction API.

5. Markov semantic property and transition representation

5.1 HasMarkovProperty is kernel-free

Canonical semantic predicate: `HasMarkovProperty X F P`. It includes `Adapted F X` and, for $s \leq t$ and measurable A , requires $E[1_{\{X_t \in A\}} | F_s] = E[1_{\{X_t \in A\}} | \sigma(X_s)]$ almost surely. The event form is the definition; the bounded-measurable test-function form is a mandatory characterization theorem via monotone-class machinery.

Do not define Markovness by bundling a chosen kernel. A chosen `TransitionSystem` / `MarkovSemigroup` is a representation and computational interface, not the semantic definition.

5.2 Conditional-independence characterization

Under the hypotheses needed by `Mathlib` conditional-independence APIs, prove the Klenke characterization: past information and future information are conditionally independent given $\sigma(X_t)$. For a general filtration use F_t and `futureMeasurableSpace X t`; for the natural filtration recover $\sigma(X_s : s \leq t) \perp \sigma(X_s : s \geq t) \mid \sigma(X_t)$. Reuse `Mathlib CondIndep/iCondIndep`; never rebuild conditional independence.

5.3 Transition representation

Define a separate relation `HasTransitionSystem X F P K`: for $r \leq s$ and measurable A , $E[1_{\{X_s \in A\}} | F_r] = K_{r,s}(X_r, A)$ a.s. The homogeneous `HasTransitionSemigroup` specialization uses κ_{s-r} only when the time type supports the legitimate difference. `HasTransitionSystem` implies `HasMarkovProperty`. Standard-Borel hypotheses are used when extracting regular conditional distributions, not in the basic Markov semantic predicate.

6. Canonical process construction and strong Markov in countable time

Markov-process existence from a `TransitionSystem`/`MarkovSemigroup` must consume the previous projective-law/Kolmogorov-extension package. Build consistent finite-dimensional laws, invoke the canonical projective extension, and use the coordinate process. Do not recreate KET inside Markov.

For discrete homogeneous chains, prefer a thin path-law specialization of `Mathlib Ionescu-Tulcea`: a canonical path kernel `markovChainPathKernel  $\kappa : \text{Kernel } E\ (\text{Nat} \rightarrow E)$` , with `markovChainPathKernel  $\kappa\ x = P_x \circ \kappa$` . Internally specialize `Kernel.traj/trajMeasure`; do not build another trajectory measure stack.

Countable-time strong Markov belongs here: ordinary Markov $\Rightarrow$ strong Markov for countable time under the exact Klenke hypotheses. Continuous-time ordinary Markov does not imply strong Markov without regularity; this is a mandatory negative semantic test.

7. Countable-state reachability, hitting, recurrence, Green function

7.1 Irreducibility ownership

Mathlib Kernel.IsIrreducible $\varphi \kappa$ is canonical. In the discrete/countable setting, specialize $\varphi = \text{Measure.count}$. Build a bridge to pairwise reachability if needed, but do not define a second irreducibility predicate. A thin $\text{Kernel.CanReach } \kappa \ x \ y$ may be added if no exact API exists: $\exists n, (\kappa^n) \ x \ \{y\} > 0$.

Important diagonal subtlety: Mathlib reachability allows $n=0$, whereas Klenke $F(x,y)$ uses the first strictly positive hitting time. Therefore the equivalence with Klenke $F(x,y)>0$ must explicitly prove the diagonal case and may not be asserted by definition.

7.2 Strict positive hitting time

Reuse Mathlib hittingAfter . For Nat time, $\text{firstPositiveHittingTime } X \ A$ is definitionally/thinly $\text{hittingAfter } X \ A \ 1$. Keep values in $\text{WithTop } \text{Nat} / \text{ENat}$ semantics; never use untopA/untopD as a substitute for finiteness. Mathlib already proves hittingAfter is a stopping time under the appropriate discrete adaptedness hypotheses.

7.3 Return probability and mean return time

Define $F_{\kappa}(x,y) = P_{x^{\kappa}}(\tau_y^{+} < \infty)$. Define $\text{meanReturnTime } \kappa \ x$ as the ENNReal expectation of the ENat -valued τ_x^{+} using the canonical $\text{ENat} \rightarrow \text{ENNReal}$ coercion. $\text{PositiveRecurrentAt } \kappa \ x$ means $\text{meanReturnTime } \kappa \ x < \infty$; RecurrentAt means $F_{\kappa}(x,x)=1$; TransientAt means $F_{\kappa}(x,x)<1$; NullRecurrentAt means recurrent but not positive recurrent.

Mandatory bridge: the tail-sum identity $m_{\kappa}(x) = \sum_{n \geq 0} P_x(\tau_x^{+} > n)$. This gives $\text{finite mean} \Rightarrow \text{a.s. finite return without arbitrary totalization}$.

7.4 Green function

Define $\text{Kernel.green } \kappa \ x \ y = \sum_n (\kappa^n) \ x \ \{y\}$ in ENNReal . Prove its occupation interpretation and Klenke recurrence criterion $G(x,x)=\infty$ iff x recurrent, then recurrence propagation along communicating states. Exercise 17.4.1-type positive-recurrence propagation is load-bearing and must be treated as a theorem, not as a disposable exercise.

8. Excursion occupation measure, invariant probability, and Kac

Canonical construction: $\mu_x = \sum_n (X_n)_* (P_{x^{\kappa}} \text{ restricted to } \{\tau_x^{+} > n\})$. This is a genuine measure-series construction using Measure.restrict , Measure.map , and countable sums. Do not reconstruct a measure from singleton weights unless Mathlib forces that specialization.

Required theorems: $\mu_x(A) = E_x[\sum_{0 \leq n < \tau_x^{+}} 1_A(X_n)]$; $\mu_x(E) = \text{meanReturnTime } \kappa \ x$; $\mu_x(\{x\}) = 1$. If x is recurrent, prove $\text{Kernel.Invariant } \kappa \ \mu_x$ by the Markov conditional-expectation shift argument and the a.s. return $X_{\{\tau_x^{+}\}} = x$.

If x is positive recurrent, normalize μ_x by its finite mass to obtain an invariant probability. Under irreducibility prove invariant-probability uniqueness and the equivalence: positive recurrence iff an invariant probability exists. Derive Kac: $\pi(\{x\}) = 1 / E_x[\tau_x^{+}]$. Do not introduce $\text{StationaryDistribution}$ or a duplicate invariant predicate; Mathlib Kernel.Invariant remains canonical.

9. Periodicity

Use kernel powers only. Define $\text{transitionTimes } \kappa \ x \ y = \{n \mid (\kappa^n) x \{y\} > 0\}$, $\text{period } \kappa \ x$ as the gcd of return times, and aperiodicity as $\text{period } 1$. Audit Mathlib for a Set Nat gcd API before building arithmetic support.

Load-bearing results are stronger than “all states have equal period”: for period d , sufficiently large admissible return times nd have positive probability; in an irreducible chain there are residues $L_{\{x,y\}}$ such that all sufficiently large $nd + L_{\{x,y\}}$ are reachable. These eventual-positivity theorems feed the product-chain irreducibility used by coalescent coupling.

10. Coupling and total variation

10.1 Generic coupling ownership

If Mathlib/local audit shows a real gap, create a generic probability $\text{IsCoupling } \gamma \ \mu \ \nu$ predicate outside Markov. It states that the two coordinate marginals of γ are μ and ν . Product measure gives an immediate witness. Do not create a Markov-specific coupling structure that bundles processes.

10.2 Hard source correction: Klenke (17.30) factor 2

Work must not copy Klenke (17.30) literally. Klenke (17.29) uses the full signed-measure TV norm: $\|P-Q\|_{TV} = \sup_{\{f \mid \|f\|_{\infty} \leq 1\}} \int f d(P-Q) = 2 \sup_A |P(A)-Q(A)|$. Under this convention the correct maximal-coupling identity is $\|P-Q\|_{TV} = 2 \inf_{\gamma} \gamma\{x \neq y\}$. The load-bearing inequality is $\|P-Q\|_{TV} \leq 2 \gamma\{x \neq y\}$. Klenke Theorem 18.12 itself uses the factor-2 inequality, confirming the normalization.

Formalize against Mathlib canonical signed-measure total variation / variation. Do not create an ambiguously normalized TVDistance. Corrected maximal-coupling equality is generic probability theory and may be BUILD-CANDIDATE if the inequality suffices for the Markov milestone, but the source correction must remain in the coverage ledger.

10.3 Successful path coupling

A Markov chain coupling is a coupling of canonical path laws P_x^κ and P_y^κ on $E^{\mathbb{N}}$. Keep Klenke Definition 18.5 as the source-faithful primary success criterion: probability of a mismatch occurring at some $m \geq n$ tends to zero as $n \rightarrow \infty$. Prove equivalence to a.s. eventual permanent equality. Pair-level HasSuccessfulCouplingFrom $\kappa \times \gamma$ may be exposed; global success is the universal quantification over x, y .

11. Independent coalescent - mandatory semantic construction, not OCR copy

Work must NOT implement Example 18.6 by copying an OCR-extracted piecewise singleton formula. The extracted text may lose “ $\neq$ ” conditions and is therefore unsafe. The correct procedure is semantic-first:

1. Define independentCoalescentKernel κ on $E \times E$.
2. If current states x and y are different, the next pair law is the product of $\kappa(x, \cdot)$ and $\kappa(y, \cdot)$.
3. If current states are equal $x=y$, draw one z from $\kappa(x, \cdot)$ and map it to (z, z) ; hence the next law is supported on the diagonal.
4. Prove this kernel is Markov and that the diagonal is absorbing.
5. Only after the semantic definition is established, derive the singleton transition piecewise formula as a theorem. Never treat the OCR piecewise cases as the source of truth.

Construct the pair-chain path law from this kernel, then push it to a coupling of two $E^{\mathbb{N}}$ paths. Prove the two marginals are exactly P_x^κ and P_y^κ . The coalescence time is the ordinary hitting time of the diagonal from time 0. Success is equivalent to a.s. finite diagonal hitting because the diagonal is absorbing.

Klenke 18.11-18.13 chain: irreducible + positive recurrent + aperiodic $\Rightarrow$ product chain irreducible; $\pi \otimes \pi$ invariant $\Rightarrow$ product chain positive recurrent $\Rightarrow$ recurrent $\Rightarrow$ diagonal hit a.s.; therefore independent coalescent is successful; therefore TV distance between evolved laws tends to zero. The final theorem must preserve Klenke 18.13's full equivalence (aperiodicity $\Leftrightarrow$ TV convergence from every state $\Leftrightarrow$ from some state $\Leftrightarrow$ from every initial probability), not weaken the target to weak convergence.

12. Continuous-time countable chains and Q-matrices

12.1 IsQMatrix - countability treatment

Adopt the general-first choice: IsQMatrix q itself is defined only by the mathematical summability/row-zero conditions and does not require [Countable E]. The actual countable-state construction theorems (uniformization kernel, sums over the state space, Klenke §17.3 process construction) add [Countable E] where it is genuinely used.

$q : E \rightarrow E \rightarrow \text{Real}$. Off-diagonal $q(x, y) \geq 0$ for $x \neq y$; off-diagonal row is summable; $q(x, x) = -\sum_{y \neq x} q(x, y)$. Use a subtype $y : \{y \mid y \neq x\}$ where convenient to avoid diagonal condition mistakes.

12.2 Q-matrix OCR safeguard

The extracted Klenke text near Theorem 17.25 may lose the “ $\neq$ ” sign and appear to say positivity for $x=y$. This is wrong. Positivity is off-diagonal $x \neq y$; the diagonal is nonpositive and equals minus the off-diagonal row sum. Work must follow the mathematical definition, not the OCR extraction.

12.3 Uniformization

Use an explicit dominating rate Λ rather than making sSup the core interface. If $\Lambda > 0$, define $P_\Lambda(x, y) = q(x, y) / \Lambda$ for $x \neq y$ and $1 + q(x, x) / \Lambda$ for $x = y$. If $\Lambda = 0$, first prove $q = 0$ and use Kernel.id; do not divide by zero and rely on Lean totalization.

Define the continuous semigroup by the Poisson mixture $K_t = \exp(-\Lambda t) \sum_n (\Lambda t)^n / n! \cdot P_\Lambda^n$. Prove it is a MarkovSemigroup using Poisson convolution and Kernel.pow_add, and prove the right derivative at zero recovers q for all x, y , including the diagonal.

Define $\text{HasQMatrix } \kappa \ q$ as the right-derivative relation on singleton transition probabilities, with safe real conversion of Markov probabilities and $t \rightarrow 0$ through positive times. Klenke 17.25 requires not only existence but uniqueness under bounded exit rates. Translate “unique Markov process” into unique transition semigroup plus unique canonical process law, never raw equality between process functions on different sample spaces. Do not replace the uniqueness proof by an unformalized “obvious $\exp(tQ)$ ” assertion.

Acceptance: the Poisson-process Q -matrix must reproduce the already existing Poisson transition law; this tests cross-package reuse.

13. Feller semigroups

13.1 Canonical definition

Use Klenke §21.26 as the canonical definition. Reuse Mathlib $C_0(E, \text{Real}) = \text{ZeroAtInftyContinuousMap}$. $\text{IsFellerSemigroup } \kappa$ requires MarkovSemigroup plus: P_t maps $C_0(E)$ into $C_0(E)$; and for every $f \in C_0(E)$, $x \in E$, $P_t f(x) \rightarrow f(x)$ as $t \downarrow 0$. Do not replace the definition by sup-norm strong continuity.

13.2 Mandatory equivalence with sup-norm strong continuity

Work must audit an existing published proof that, under the appropriate locally compact / Feller hypotheses, the Klenke pointwise-at-zero definition is equivalent to the common C_0 sup-norm strong-continuity formulation $\|P_t f - f\|_\infty \rightarrow 0$. Keep only one Feller predicate (the Klenke one); expose strong continuity as a theorem/characterization.

Concrete proof pointers for search/audit: Klenke §21.26 for the project definition; Revuz & Yor, Continuous Martingales and Brownian Motion, Chapter III, Proposition 2.4 is a standard cited pointer for the Feller criterion/equivalence used in the literature. If the exact proposition text is publicly accessible, audit it and formalize the verified argument. If it cannot be accessed in sufficient detail, mark this bridge DEFER and do not independently invent a proof. A common proof strategy uses the C_0 contraction, resolvents $R_\lambda f = \int_0^\infty e^{-\lambda t} P_t f \, dt$, density of the resolvent range, strong continuity on the dense range, and contraction to extend to all C_0 .

This theorem is valuable because it creates a machine-checked bridge between Klenke’s probability formulation and the operator-semigroup formulation used by later generator theory and some external Feller-process theorems.

13.3 Operator layer

After C_0 preservation, define the derived Feller operator P_t on $C_0(E)$. Prove contraction $\|P_t f\|_\infty \leq \|f\|_\infty$ and operator semigroup law $P_{s+t} = P_s \circ P_t$. The order is consistent with kernel orientation $\kappa_{s+t} = \kappa_t \circ \kappa_s$.

Use the previous measure-convergence package to derive fixed-time weak continuity $x_n \rightarrow x \Rightarrow \kappa_t(x_n, \cdot) \Rightarrow \kappa_t(x, \cdot)$, and stochastic continuity $\kappa_t(x, \cdot) \Rightarrow \delta_x$ as $t \downarrow 0$. Do not reprove Portmanteau or weak/vague conversion here.

14. Klenke 21.27: Feller $\Rightarrow$ RCLL strong-Markov realization

14.1 Status changed: external-proof audit then formalize

Klenke does not provide the full proof. Work must follow Klenke’s explicit external references, verify the exact result and hypotheses, and only then formalize the audited proof into StochLean. Klenke’s discussion points to Rogers & Williams, Diffusions, Markov Processes and Martingales, Volume 1, Chapter III, §7ff for the common RCLL realization construction and §8ff for strong Markov; it also points to Revuz & Yor, Continuous Martingales and Brownian Motion, Chapter III, Theorem 2.7 for the strong-Markov/Feller-process route. Search these exact references first.

External-proof rule: if the relevant source text is publicly accessible, audit theorem statement, assumptions, and proof dependencies. If correct, formalize the proof and only the missing prerequisite closure under their canonical owners. If the specific source content is not accessible, record MK-B05 as DEFER with the exact bibliographic pointer and skip the theorem. Do not attempt an independent proof.

14.2 Expected formalization decomposition

Prefer staged theorems rather than one giant proof: (A) Feller semigroup $\Rightarrow$ canonical Markov coordinate-process law using prior KET; (B) existence of an RCLL/càdlàg modification compatible across initial states; (C) identify/construct the right-continuous completed/usual information flow actually used by the audited proof; (D) prove strong Markov for appropriate a.s.-

finite stopping times. Final conclusion is existence of an RCLL strong-Markov realization with transition semigroup κ , not a claim that every realization is RCLL.

Do not duplicate generic usual augmentation, Doob regularization, càdlàg machinery, or random-time measurability inside Markov. If the audited external proof needs them, create them under the previously assigned canonical owners and record the dependency in the blocker ledger.

15. Source correction and audit register

Source location	Issue / interpretation	Normative action
Klenke §14.42-14.43	Probability kernel multiplication order differs from Mathlib $\circ_k$	Use Lean law $K \circ r \circ t = K \circ s \circ t \circ_k K \circ r \circ s$; repeat in Work report
Klenke Ex. 18.6	OCR may corrupt piecewise conditions (especially $\neq$)	Define independent coalescent semantically first; derive piecewise formula afterward
Klenke (17.30)	Printed coupling equality misses factor 2 under the book's own full TV norm	Use $\ \mu - \nu\ _{TV} = 2 \inf \text{mismatch}$ and $TV \leq 2 \text{ mismatch}$
Klenke §17.3 / Thm 17.25 extraction	OCR may drop $\neq$ in off-diagonal positivity	Require $q(x,y) \geq 0$ only for $x \neq y$; diagonal is negative row sum
Klenke §21.26	Pointwise C_0 -at-zero Feller definition vs common strong C_0 continuity	Keep Klenke definition; audit/formalize equivalence theorem
Klenke 21.27	Full proof delegated to external literature	Audit Rogers-Williams III.7ff/III.8ff or Revuz-Yor III Thm 2.7; formalize if accessible, else DEFER

16. Functional module boundary

- Probability/
 - Process/
 - Filtration/
 - Future.lean
- Markov/
 - Basic.lean
 - TransitionSystem.lean
 - Semigroup.lean
 - Transition.lean
 - StrongMarkov.lean
 - Feller.lean
- Chain/
 - PathLaw.lean
- Countable/
 - Hitting.lean
 - Recurrence.lean
 - Green.lean
 - Excursion.lean
 - Invariant.lean
 - Periodicity.lean
 - Coalescent.lean
 - Convergence.lean
- Generator/
 - QMatrix.lean
 - Uniformization.lean
- Coupling/
 - Basic.lean
 - TotalVariation.lean

This is an ownership boundary, not a required file checklist. If duplicate audit closes a gap, omit the local file. Do not mirror Mathlib Kernel, Invariant, Irreducible, Reversible, HittingTime, Portmanteau, or Ionescu-Tulcea modules.

17. Work implementation order

Phase	Required work
A	Repeat duplicate/API audit: pinned Mathlib → complete local StochLean → approved external ledger. Resolve inherited KET blocker before process construction.
B	Build/reuse futureMeasurableSpace and information-flow theorems.
C	Build TransitionSystem first; then MarkovSemigroup and homogeneous bridges. Lock orientation tests.
D	Build HasMarkovProperty, bounded-function and conditional-independence characterizations, HasTransitionSystem, and canonical Markov construction.
E	Build canonical chain path kernel via Ionescu-Tulcea specialization; strict positive hitting, $F(x,y)$, mean return, Green, recurrence propagation.
F	Build excursion occupation measure, invariance, normalization, irreducible invariant-probability uniqueness, positive-recurrence equivalence, Kac formula.
G	Build periodicity and eventual positivity/residue theorems.
H	Build generic coupling and corrected TV inequality; semantic independent coalescent; Klenke 18.11-18.13 full TV convergence equivalence.
I	Build IsQMatrix, bounded-exit uniformization, Poisson mixture semigroup, HasQMatrix recovery, and uniqueness; run Poisson acceptance.
J	Build Feller definition/operator/weak-continuity layer. Audit and, if accessible, formalize pointwise $\Rightarrow$ sup-norm strong-continuity proof.
K	Audit Klenke 21.27 external references. If accessible and correct, formalize the RCLL strong-Markov realization using canonical-owner prerequisites. If inaccessible, mark DEFER and skip independent proof.
L	Final coverage, source-correction, blocker/deferred ledger, semantic regression, axiom/dependency, minimal imports, lint, no sorry/admit/axiom, and Work push report.

18. Mandatory semantic regression suite

- futureMeasurableSpace is antitone and its discrete intersection agrees with the existing tail sigma-field.
- TransitionSystem is defined only on $r \leq s$; no meaningless backward kernels are totalized.
- Kernel composition orientation tests explicitly verify $K \circ r \circ t = K \circ s \circ t \circ_k K \circ r \circ s$.
- MarkovSemigroup discrete specialization agrees with Kernel.pow_add.
- HasMarkovProperty is kernel-free; bounded measurable and conditional-independence formulations are proved bridges.
- Ordinary continuous-time Markov does not automatically imply strong Markov.
- Kernel.IsIrreducible Measure.count remains the only irreducibility notion; zero-time reachability is not confused with Klenke strict return probability.
- firstPositiveHittingTime uses hittingAfter ... 1 and preserves infinity; no arbitrary untop fallback.
- Finite mean return implies a.s.-finite strict return; excursion occupation mass equals mean return and atom at start equals 1.
- Positive recurrence $\Leftrightarrow$ invariant probability under irreducibility; invariant probability is unique; Kac formula holds.
- Periodicity tests include eventual admissible-time positivity, not merely equal periods.
- Klenke (17.30) factor-2 correction is enforced by a normalization regression.
- Independent coalescent is constructed from semantic product/diagonal cases; no OCR piecewise formula is used as a definition.
- Successful coupling uses Klenke tail-mismatch criterion and is equivalent to a.s. eventual equality.
- Klenke 18.13 target is total-variation convergence equivalence; weak convergence is only a corollary.
- IsQMatrix off-diagonal positivity and diagonal negative row sum survive OCR safeguards.
- $\Lambda=0$ uniformization proves $q=0$ and returns identity; no division-by-zero totalization.
- Uniformized semigroup recovers q on both diagonal and off-diagonal derivatives; uniqueness is proved, not asserted by $\exp(tQ)$.
- Poisson Q-matrix reproduces the pre-existing Poisson package law.
- Feller uses Mathlib C_0 and Klenke pointwise definition; no second C_0 type or second Feller predicate.
- If the strong-continuity equivalence is formalized, both directions are tested and external source/proof provenance is recorded.
- Feller operator is contractive and $P_{\{s+t\}} = P_s \circ P_t$ in operator order.
- Feller fixed-time transition laws are weakly continuous in the state and converge to δ_x as $t \downarrow 0$.

- Klenke 21.27 external theorem is formalized only after exact source audit; inaccessible source yields DEFER, not an autonomous proof attempt.
- RCLL strong-Markov result is an existence/version theorem, not a claim about every realization.
- Final blocker/deferred ledger has no unlabeled TODO and every row is RESOLVED, STILL-BLOCKED, or DEFERRED with owner and reason.

19. Completion gate

The package is complete only when all BUILD items in the chosen current scope are proved, every AUDIT-FIRST/EXTERNAL-PROOF item is resolved or explicitly deferred under the external-proof protocol, all source corrections are encoded in the Work report, and the blocker/deferred ledger is complete. No sorry, admit, custom axioms, vacuous assumptions, semantic weakening, or hidden totalization is permitted. Work must inspect theorem axioms/dependencies, run lint/minimal-import checks, and confirm that no previous package was duplicated.

The Markov package may still ship with explicitly DEFERRED items whose source text is inaccessible under the new protocol, provided they are not silently required by a theorem claimed complete and their downstream blockers are recorded.